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Analytic relationship of relative synchronizability to network structure and motifs

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Synchronization phenomena on networks have attracted much attention in studies of neural, social, economic, and biological systems, yet we still lack a systematic understanding of how relative synchronizability relates to underlying network structure. Indeed, this question is of central importance to the key theme of how dynamics on networks relate to their structure more generally. We present an analytic technique to directly measure the relative synchronizability of noise-driven time-series processes on networks, in terms of the directed network structure. We consider both discrete-time autoregressive processes and continuous-time Ornstein–Uhlenbeck dynamics on networks, which can represent linearizations of nonlinear systems. Our technique builds on computation of the network covariance matrix in the space orthogonal to the synchronized state, enabling it to be more general than previous work in not requiring either symmetric (undirected) or diagonalizable connectivity matrices and allowing arbitrary self-link weights. More importantly, our approach quantifies the relative synchronization specifically in terms of the contribution of process motif (walk) structures. We demonstrate that in general the relative abundance of process motifs with convergent directed walks (including feedback and feedforward loops) hinders synchronizability. We also reveal subtle differences between the motifs involved for discrete or continuous-time dynamics. Our insights analytically explain several known general results regarding synchronizability of networks, including that small-world and regular networks are less synchronizable than random networks.

synchronization | complex networks | motifs

Significance

How does the quality of synchronization between coupled oscillators depend on the network structure that connects them? This question has been at the forefont of studies on the structure–dynamics relationship, one of the most important open problems in complex systems. We present a method to fully and generally relate the structure of a network to its synchronizability, without assumptions made by previous approaches such as symmetric connections. Moreover, we reveal the impact of patterns of network connections among small groups of nodes (motifs) on the whole of network synchronizability. Our results implicate the prevalence of clustered structure such as feedforward and feedback loops as the most important factor in synchronizability.

The manner in which systems of coupled oscillators achieve synchronization is one of the most widely occurring, compelling, and oldest examples analyzed in complex systems science (1–3). Synchronization has been observed and studied in many different and seemingly unrelated fields, including in swarms of flashing fireflies (4), neural synchrony (5–7), population dynamics in biology (8), computer systems (9), and many others. While much early study focused on the Kuramoto model (10, 11) with full coupling between the oscillators (or via mean field interactions), the study of synchronization is naturally generalized to complex networks of interaction between the oscillators (e.g., ref. 1 and references therein). In this setting, each oscillator adjusts its activity in a decentralized fashion, as a function of the activity of the subset of oscillators that it is connected to via the complex network. While the capacity for synchronization of the oscillators, and the relative quality of such synchronization, varies with the strength of connectivity, it will clearly also vary with the underlying network structure (1, 12). Establishing a full analytic relationship between network structure and synchronizability is our focus.

Indeed, this relates to the more general research question of how dynamics on such complex networks relate to their structure (12). While the importance of such investigations has long been recognized, characterizing the structure–dynamics relationship remains "much less well understood" than characterizing structure alone (13) since the problem is simply more diverse (14). How synchronization relates to network structure between oscillators is seen to be of central importance to this wider structure–dynamics question (1, 12) because of the significant interdisciplinary and historical interest in the dynamic process of synchronization.

Formally, given a directed network structure and coupling strengths, we aim to determine whether the network is synchronizable and then to rank the relative quality of that synchronization (in terms of robustness to perturbations away from a synchronized state). As described in Section 1 summarizing (1, 2, 15–20), we will take the standard approach of examining linear systems, which can be considered as a linearization around attractors of nonlinear systems, where general analytic insights are possible. In this regime, one can easily write down synchronization conditions in terms of the eigenvalues of the network coupling matrix or weighted adjacency matrix C (or equivalently those of the

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system Laplacian L) (17, 19, 20). But since for linear systems, broad ranges of coupling strengths can guarantee synchronization for any network structure (17), it is useful to have insights into the relative synchronizability of a network beyond simply answering whether it will synchronize or not (20). Relative synchronizability can be quantified as the expected steady-state distance from synchronization $\langle\sigma^2\rangle$ under noise-driven dynamics (or equivalently for deterministic dynamics, the expected total integrated/summed deviation from synchronization caused by an initial perturbation; see Section 1). This has been computed analytically using all eigenvectors and all eigenvalues for symmetric C (18–20). Heuristics based on leading eigenvalues or spread of eigenvalues are also often used (16, 20–22).

Yet despite the analytic success under these conditions, it could not be said that we have a systematic analytic understanding of how relative synchronizability relates to network structure. This is because, apart from the extremal eigenvalues which are related to global properties (23, 24), there is no known general method to relate eigenvalues of the coupling matrix C or the Laplacian L to network structure. Specifics are known only for very special cases, such as fully connected and bipartite graphs, yet for nonsymmetric networks, the corresponding coupling operator is not self-adjoint anymore, and only in very special cases connections between eigenvalues and structure of the network are known (25). There are of course many studies empirically evaluating the dependence of synchronizability on network structure either based on the above analytic forms from the eigenvalue spectra (19, 20) or otherwise heuristic approximations (16, 20–22, 26–29), numerical simulations (30–35), or evaluation of the synchronizability of standalone motif structures outside of a network setting (36–38). Some results across these are relatively consistent, e.g., that more random network structures tend to synchronize more effectively than regular ring or small-world structures (1, 9, 26, 28, 30–32, 39) and that degree homogeneity is helpful for synchronization (1, 20, 21). However, Arenas et al. (1) state that due to difficulties in isolating single network characteristics while controlling others, results can be conflicting, meaning that the situation remains "quite confusing" and can be misinterpreted. Crucially, even where we can analytically compute a measure of synchronizability of a network, we cannot yet fully articulate the contribution of local subnetwork motif structures (40) such as feedback loops to promoting or detracting from whole of network synchronizability.

Here, we provide a full and general analytical solution to this problem, including quantifying how the different network motifs shape the synchronizability of the network as a whole. Specifically, in Sections 2 and 3, we introduce a method to measure the relative synchronizability via the steady-state distance from synchronization $\langle\sigma^2\rangle$ of a network structure (with final results in Eqs. **22** and **23**). In comparison to previous analytic work (19, 20), our method determines relative synchronizability without requiring symmetry, nonnegativity, or diagonalizability of the weighted coupling matrix C. This is important theoretically in view of generality of results, e.g., the importance of diagonalizability (41), but particularly, in view of biological applications, for instance, neuronal network models which are usually directed and use positive and negative weights corresponding to excitatory and inhibitory synapses, with these playing different roles in synchronization (21). More importantly, our approach naturally provides an analytic description relating relative synchronizability to local network structures, specifically "process motifs" as structured sets of walks on subnetwork structures (42). We will show in Section 4 that networks become less synchronizable as the (weighted) sum of paired walks from each node that converge to a common endpoint rather than diverge becomes larger. We show subtle differences between discrete and continuous-time dynamics, where with discrete-time dynamics, only convergent walks of the same length detract from synchronizability, whereas in continuous-time dynamics, more general feedforward and feedback process motif structures also detract. These results have important implications for our understanding of natural networks and their synchronizability, in particular regarding synchronization phenomena in neural and cell-regulatory networks which exhibit a prevalence of such loop process motif structures (40), in addition to directed and signed edge weights. We expect that the analytic forms we provide will be utilized to analytically explore a wide range of new relationships between network structure and relative synchronizability.

1. Methods: Relative Synchronizability

A. Linear(ized) Models of Dynamics on a Network Structure. Given the structure and edge weights C of a network of N nodes, we consider the dynamics on such networks in two very simple and widely used models of linearly coupled noise-driven systems. These are the following: i. the continuous-time (differential equation) multivariate Ornstein–Uhlenbeck process (42–45) also known as the Edwards–Wilkinson (46) process on a network (19, 20):

$$d\vec{x}(t) = -\vec{x}(t)(I - C)\theta dt + \zeta d\vec{w}(t), \qquad [1]$$

and ii. the discrete-time (difference equation) multivariate or vector autoregressive (VAR) process (45):

$$\vec{x}(t+1) = \vec{x}(t)C + \zeta\vec{r}(t), \qquad [2]$$

on that network. In both cases, $C = [C_{ji}]$ is the $N \times N$ connectivity matrix (or weighted adjacency matrix), where C_{ji} is the (real, possibly negative) weight of the directed connection from node j to i, and I is the $N \times N$ identity matrix. The current node values or activity levels $\vec{x}(t) = \{x_1(t), x_2(t), \ldots x_N(t)\}$ is a row vector here.

These processes, with reversion rate $\theta > 0$ in continuous-time, are driven by uncorrelated noise terms with strength ζ^2 (for a multivariate Wiener process $\vec{w}(t)$, and mean-zero unit-variance Gaussian noise $\vec{r}(t)$) with covariance matrix equal to $\zeta^2 I$ in both cases. These noisy perturbations put the network into a perennial transient with respect to the synchronized state or zero mode $\vec{\psi}_0 = [1, 1, \ldots, 1]$ as a potential attractor. Indeed, most studies (e.g., refs. 18–20, and 39) consider C with $\vec{\psi}_0$ as an eigenvector with eigenvalue $\lambda_0 = 1$ (so the sum of all incoming edges to a target $\sum_j C_{ji} = 1$, i.e., with diffusive coupling), giving the possibility of a dynamic synchronization around $\vec{\psi}_0$ rather than trivial synchronization around the null state. We will consider C both with and without $\vec{\psi}_0$ as an eigenvector. We also note that other studies often focus on the Laplacian instead of the coupling matrix C; specifically, here, the normalized Laplacian (47) $L = I - C$ (when $\vec{\psi}_0$ is an eigenvector with eigenvalue $\lambda_0 = 1$).

The approach of examining linear dynamics can also be broadly interpreted as considering a linearization around a potentially nonhomogeneous attractor for nonlinear dynamics, assuming that the system stays within its basin of attraction (see refs. 1, 2, 15–20 and references therein). *SI Appendix*, section 1 [following (1, section 4)] describes the use of the Master Stability Function

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approach (48, 49) to consider linearizations of continuous-time systems of general forms including either of

$$\frac{dy_i(t)}{dt} = f(y_i(t)) + \sum_{j=1}^{N} C_{ji}(h(y_j(t)) - h(y_i(t))) + \zeta \frac{dw_i(t)}{dt},$$
$$\frac{dy_i(t)}{dt} = f(y_i(t)) + \sum_{j=1}^{N} C_{ji} h(y_j(t) - y_i(t)) + \zeta \frac{dw_i(t)}{dt}, \quad [3]$$

where f and h may be nonlinear mappings. The linearization of fluctuations $\vec{x}(t)$ around a synchronization solution of $\vec{y}(t)$ results in dynamics for $\vec{x}(t)$ consistent with Eq. **1**. Linear stability of the synchronized state here is a necessary condition for synchronization of the nonlinear system (1). This approach has been used to study synchronization of the Kuramoto model (50, 51), Rössler dynamics (1), and for discrete-time coupled map lattices in ref. 16 with application to the logistic map.

B. Synchronization Conditions. It is well known that the necessary synchronization conditions (for the case without constant driving noise; demonstrated for constant driving noise in ref. 19) can be stated in terms of the eigenvalues λ_v of C (or translated to those of L):

1. For continuous-time dynamics in Eq. **1**, we require $\mathrm{Re}(\lambda_v) < 1, \forall \lambda_v$ (except for λ_0 when $\vec{\psi}_0$ is an eigenvector).
2. For discrete-time dynamics in Eq. **2**, we require $|\lambda_v| < 1, \forall \lambda_v$ (again except for λ_0 when $\vec{\psi}_0$ is an eigenvector).

Since these conditions can be met for any network structure by multiplicatively lowering the magnitudes of the C_{ji} where required (17)*, it is useful to have insights into relative synchronizability beyond simply answering whether the network will synchronize or not (20). For various types of dynamics (not necessarily linear systems, with or without time delays), one can make inferences regarding the relative synchronizability of various networks from their eigenvalue spectra (52) (ignoring λ_0 corresponding to $\vec{\psi}_0$). While the precise definition of synchronizability differs with dynamics, the general picture is that the relative synchronizability increases with the distance of the eigenvalues (in particular the largest) from the stability boundaries above and also as the spread of eigenvalues becomes smaller (16, 20–22, 52).

C. Steady-State Distance from Synchronization $\langle\sigma^2\rangle$. Toward a more precise alternative, we can compute

$$\sigma^2(t) = \frac{1}{N}\sum_{i=1}^{N}(x_i(t) - \bar{x}(t))^2, \quad [4]$$

to measure the distance from synchronization at time t, with $\bar{x}(t) = \frac{1}{N}\sum_{i=1}^{N} x_i(t)$. Then, to evaluate relative synchronizability directly in the noise-driven cases of Eqs. **1** and **2**, we compute $\langle\sigma^2\rangle$ (9, 20, 39):

$$\langle\sigma^2\rangle = \lim_{t\to\infty}\frac{1}{N}\sum_{i=1}^{N}\left\langle(x_i(t) - \bar{x}(t))^2\right\rangle, \quad [5]$$

the expectation value which "represents an average over the statistical ensemble" (45) over the noise (18), dropping the time argument to indicate steady-state (20). Under the standard synchronization conditions (above), this average steady-state distance from the synchronized state (or "width of the synchronization landscape") "approaches a finite value in the $t \to \infty$ limit" (18), i.e., $\langle\sigma^2\rangle < \infty$ (18, 19), in contrast to $\langle\sigma^2\rangle$ diverging when the synchronization conditions are not met.† So, since $\langle\sigma^2\rangle$ takes a range of finite values when the network satisfies the synchronization condition (20), it provides a nuanced and meaningful indication of relative synchronizability, with smaller values indicating stronger relative synchronizability under driving noise. For practical use, one could compute $\langle\sigma^2(t)\rangle$ empirically via Eq. **4** after some fixed t (e.g., ref. 39); however, we will show that analytically evaluating $\langle\sigma^2\rangle$ yields further insights.

A common alternative is to consider the response in deterministic dynamics to a single perturbation, i.e., to set $\vec{x}(0) = \zeta\vec{r}(0)$ as the initial condition in both Eqs. **1** and **2**, and then set $\vec{r}(t) = 0$ and $d\vec{w}(t) = 0$ for $t > 0$. Here, one can define the expected total deviation from synchronization due to a single perturbation, integrating $\sigma^2(t)$ from Eq. **4** over all subsequent time-steps after the perturbation for continuous time:

$$D = \left\langle\int_0^\infty \sigma^2(t)dt\right\rangle, \quad [6]$$

or summed over all time steps for discrete time:

$$D = \left\langle\sum_{t=0}^{\infty}\sigma^2(t)\right\rangle, \quad [7]$$

where the expectation is over the ensemble of noise $\vec{r}(0)$. Under the synchronization condition in both cases, $\sigma^2(t) \to 0$ as $t \to \infty$ and we show in *SI Appendix*, section 6 that $D = \langle\sigma^2\rangle$, i.e., the total integrated/summed deviation from the synchronized state in the deterministic case is precisely equivalent to the expected steady-state deviation from sync in the driving noise case. As such, we focus on $\langle\sigma^2\rangle$ for the remainder.

We observe that

$$\langle\sigma^2\rangle = \lim_{t\to\infty}\frac{1}{N}\mathrm{trace}\left(\left\langle(U^T\vec{x}(t)^T)(\vec{x}(t)U)\right\rangle\right), \quad [8]$$
$$= \frac{1}{N}\mathrm{trace}\left(U^T\Omega U\right),$$
$$= \frac{1}{N}\mathrm{trace}\left(\Omega_U\right), \quad [9]$$

where U is the centering matrix or unaveraging operator‡ $\vec{x}U = \vec{x} - \bar{x}\vec{\psi}_0$ (53); while $\Omega = \lim_{t\to\infty}\langle\vec{x}(t)^T\vec{x}(t)\rangle$ (with elements $\Omega_{ij} = \lim_{t\to\infty}\langle x_i(t)x_j(t)\rangle$) is the steady-state symmetric covariance matrix of $\vec{x}(t)$; and $\Omega_U \triangleq \lim_{t\to\infty}\langle U^T\vec{x}(t)^T\vec{x}(t)U\rangle$, and $\Omega_U = U^T\Omega U$ when Ω is well-defined. Note that U makes projections onto the space orthogonal to the zero-mode $\vec{\psi}_0$, so

*Though in lowering λ_0 of $\vec{\psi}_0$, this may result in trivial synchronization around the null state.

†In the absence of driving noise in Eqs. **1** and **2**, the synchronization condition implies that $\sigma^2(t) \to 0$ as $t \to \infty$ (39). This means that the network would completely synchronize to a common state without driving noise. In other words, the synchronization condition is that the synchronized states are *stable*; i.e., the system is stable to disturbances orthogonal to $\vec{\psi}_0$ (20). In the presence of driving noise, then, $\langle\sigma^2\rangle < \infty$ implies that the network will maintain a perennial approach to synchronization without diverging.

‡See further definition and useful expressions in *SI Appendix*, section 2, including that $U^T = U$.

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we can interpret Eq. **9** as measuring $\langle\sigma^2\rangle$ proportional to the trace of Ω in the space orthogonal to $\vec{\psi}_0$.

To analytically evaluate $\langle\sigma^2\rangle$, then, we must determine Ω_U or trace (Ω_U) directly from C (or its eigenvalues).

D. Existing Methods to Determine $\langle\sigma^2\rangle$. Galan (54) presents an eigenvalue decomposition technique for computing Ω for a generalized case of Eq. **1** directly from the eigenvectors and eigenvalues of C.[§] However, as presented, this is not applicable to C in the common case (for synchronization) when it has $\lambda_0 = 1$ for eigenvector $\vec{\psi}_0$ and requires diagonalizable C.

Korniss (20) computes $\langle\sigma^2\rangle$ for a continuous-time system (akin to Eq. **1**) using a Green's function approach to the covariance components. (When mapped to our Eq. **1**), this approach requires C to be symmetric, requires positive edge weights $C_{ij} > 0$, and assumes the common case that $\vec{\psi}_0$ is an eigenvector of C with $\lambda_0 = 1$. Using the well-known eigenvalue expansion for the Green's function (or two-point correlation function) $\langle\sigma^2\rangle$ is computed for such C with $\theta = \zeta = 1$ as[¶]

$$\langle\sigma^2\rangle = \frac{1}{2N}\sum_{v=1}^{N-1}\frac{1}{1-\lambda_v}, \qquad [10]$$

where λ_v are the eigenvalues of C except for λ_0. Insights were then derived into specific network structures (20)—i.e., for fixed total edge costs, a fully connected network of identical edge-weights minimizes $\langle\sigma^2\rangle$; while for fixed edge weights and a fixed average degree, a perfectly homogeneous random graph where each node connects to exactly k others is optimal. Hunt et al. (19) extended these results to compute $\langle\sigma^2\rangle$ for a coupling delay.

2. Results: Analytic Determination of Steady-State Distance from Synchronization $\langle\sigma^2\rangle$

In this section, we first describe a method to compute the projected covariance matrix Ω_U for Eqs. **1** and **2** directly from C in Section 2.A. Since this method, unlike previous work, does not require either symmetric or diagonalizable C, we are thus able to compute the steady-state distance from synchronization $\langle\sigma^2\rangle$ without such restrictions in Section 2.B. Numerical validation is provided in Section 2.C. Perhaps more importantly than the result for $\langle\sigma^2\rangle$ itself though, our method paves the way to directly reveal the contribution of motif structures on $\langle\sigma^2\rangle$ in the following Section 3.

A. Power Series Method for Ω_U. First, we contribute an approach to computing Ω_U, which does not require symmetry or diagonalizability of C. This is based on Barnett et al.'s method (45) to analytically obtain Ω as a power series of C.[#]

For the continuous-time process with $\theta = \zeta = 1$ in Eq. **1**, Barnett et al. (45) show that the covariance matrix Ω is obtained via a power series (45):

$$2\Omega = I + \frac{1}{2}(C^T + C) + \frac{1}{4}\left[(C^2)^T + 2C^TC + C^2\right] + \ldots,$$
$$= \sum_{m=0}^{\infty} 2^{-m}\sum_{u=0}^{m}\binom{m}{u}(C^u)^T C^{m-u}. \qquad [11]$$

The validity of this solution relies on both stationarity of Eq. **1** (given by $\mathrm{Re}(\lambda) < 1$, for all real parts of eigenvalues λ of C) and convergence of Eq. **11** (shown only for the stronger condition of $\rho(C) < 1$ for the spectral radius of C in ref. 45).

As such, this solution does not directly apply for our general case with $\lambda_0 = 1$ for the zero-mode $\vec{\psi}_0$, which is important for nontrivial synchronized solutions.[‖] To address this, we have extended the derivations of Barnett et al. (45) and Schwarze and Porter (42) in this case to examine Ω_U (in the space orthogonal to the zero-mode), demonstrating in *SI Appendix*, section 4 (c.f. *SI Appendix*, Eq. **54**):

$$\Omega_U = \frac{\zeta^2}{2\theta}\sum_{m=0}^{\infty} 2^{-m}\sum_{u=0}^{m}\binom{m}{u}U(C^u)^T C^{m-u}U, \qquad [12]$$

and showing the validity of this solution for $\rho(CU) < 1$, i.e., for $|\lambda_C| < 1$ for all eigenvalues λ_C of C except that corresponding to $\vec{\psi}_0$. Clearly, this matches $\Omega_U = U^T\Omega U$ with $\theta = \zeta = 1$ obtained directly from Ω in Eq. **11** for the more restricted regime where Eq. **11** is valid. Intuitively, we can view this as a projection of the covariance matrix onto the space orthogonal to the synchronized state $\vec{\psi}_0$. We emphasize though (as per footnote ‖) that one cannot simply write this down when Eq. **11** is not valid; the foundation provided by the proof in *SI Appendix*, section 4 is crucial.

For the discrete-time process with $\zeta = 1$ in Eq. **2**, Barnett et al. show the general solution for Ω as

$$\Omega = I + C^TC + (C^2)^TC^2 + \ldots = \sum_{u=0}^{\infty}(C^u)^TC^u. \qquad [13]$$

Again, the validity of this solution requires stationarity of Eq. **2** and convergence of Eq. **13** (given by $|\lambda_C| < 1$ for both).

As such, this solution does not directly apply for our general case with $\lambda_0 = 1$ for the zero-mode $\vec{\psi}_0$. To address this, we again extend the derivation in ref. 45 to examine Ω_U, demonstrating in *SI Appendix*, section 4 (c.f. *SI Appendix*, Eq. **65**):

$$\Omega_U = \zeta^2\sum_{u=0}^{\infty}U(C^u)^TC^uU, \qquad [14]$$

and showing there the validity of this solution for $\rho(CU) < 1$. Again, this clearly matches $\Omega_U = U^T\Omega U$ with $\zeta = 1$ obtained directly from Ω in Eq. **13** for the more restricted regime where Eq. **13** is valid.

B. Solution for Steady-State Distance from Synchronization $\langle\sigma^2\rangle$. With the above solutions for Ω_U, we can now provide the solutions for $\langle\sigma^2\rangle$.

If we have $|\lambda_C| < 1$, for all eigenvalues λ_C of C, then both Eqs. **11** and **13** are valid, and we can immediately evaluate Eq. **9** with Ω to get $\langle\sigma^2\rangle$. Alternatively, in the more interesting case with

[§]These calculations were made for a more general case of our continuous-time system Eq. **1** with correlated noise. However, the solution presented is actually for a discretization corresponding to Eq. **2**, which no longer corresponds to the continuous-time equation [as shown by Barnett et al. (45)].

[¶]We have mapped the solution to eigenvalues of C rather than L, and for noise $\vec{w}(t)$ with the same covariance we consider here [Korniss (20) considered noise with twice as much covariance].

[#]Ref. 45 corrects an original attempt by Tononi et al. (55). Similar methods are also presented in refs. 56 and 57.

[‖]One cannot trivially compute $\Omega_U = U^T\Omega U$ from Eq. **11** by relying on nonexistent convergence of Ω.

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the zero-mode $\vec{\psi}_0$ with $\lambda_0 = 1$ (where nontrivial synchronized solutions are possible), then with $|\lambda_C| < 1$ for all eigenvalues λ_C of C except that corresponding to $\vec{\psi}_0$, Eqs. **12** and **14** are valid, and we can evaluate Eq. **9** from these. Both approaches lead to identical forms for Ω_U and $\langle\sigma^2\rangle$.

As such, for the continuous-time process in Eq. **1**, we expand Eq. **9** using Eq. **12** to obtain the solution for steady-state distance from sync:

$$\langle\sigma^2\rangle = \frac{\zeta^2}{2\theta}\sum_{m=0}^{\infty}\frac{2^{-m}}{N}\sum_{u=0}^{m}\binom{m}{u}\text{trace}\left(U(C^u)^T C^{m-u}U\right). \quad [15]$$

Similarly, for the discrete-time process in Eq. **2**, we expand Eq. **9** using Eq. **14** to obtain the corresponding solution for steady-state distance from sync:

$$\langle\sigma^2\rangle = \frac{\zeta^2}{N}\sum_{u=0}^{\infty}\text{trace}\left(U(C^u)^T C^u U\right). \quad [16]$$

These solutions represent a substantial advance over existing methods, in neither requiring symmetric nor diagonalizable C, and we are thus able to compute the steady-state distance from synchronization $\langle\sigma^2\rangle$ without such restrictions. Strictly, we do require $|\lambda_C| < 1$ for all eigenvalues λ_C of C except that corresponding to $\vec{\psi}_0$, though this is equivalent to the synchronizability condition in discrete time (Section 1.B).

C. Numerical Validation. With the analytic solutions in place in Eqs. **15** and **16**, we seek to provide numerical validation for them. Matlab code for these experiments is distributed in the *linsync* toolbox (58), including adaptations of *n_comp* (59).

In order to check the validity of the solution over a wide variety of networks, we generate sample network structures throughout a small-world transition on an N-node Watts–Strogatz ring network (60). Initially, each node has d directed incoming edges from its closest neighbors on the ring, and then sources of each edge are randomized with probability p. The d incoming edges for each node i have equal edge weights $C_{ji} = c/d$, with total incoming (nonself) connection weights $c = \sum_{j\neq i} C_{ji}$, and each node has self-link weights $C_{ii} = 1 - c$ (which are not randomized). This setup with $\sum_j C_{ji} = 1$ ensures that each connectivity matrix C has zero-mode eigenvector ψ_0 with $\lambda_0 = 1$, facilitating dynamic synchronization around ψ_0. In the few cases where the edge randomization process results in networks which do not meet the requirement of $|\lambda_C| < 1$ for all other eigenvalues λ_C (and by implication, the synchronization conditions in Section 1.B), e.g., by disconnecting the network, these network realizations are discarded.

Numerical results are generated for the continuous-time dynamics in Eq. **1** with $\theta = \zeta = 1$. We compute the steady-state distance from synchronization $\langle\sigma^2\rangle$ analytically via Eq. **15**. We also compute this quantity empirically from time series simulated from Eq. **1** using the exact method following Appendix A of Barnett et al. (45) with samples taken every $dt = 1.0$ time units and thus compute the average $\langle\sigma^2\rangle_E$ over L time-series samples via Eq. **4**.

Fig. 1 plots the mean of the relative absolute error $\left|\langle\sigma^2\rangle - \langle\sigma^2\rangle_E\right| / \langle\sigma^2\rangle$ for parameters $N = 100$, $d = 4$, and $c = 0.5$, over various values of network randomization parameter p, and various sampling lengths L. For each parameter combination, the mean is taken over 2,000 network realizations.

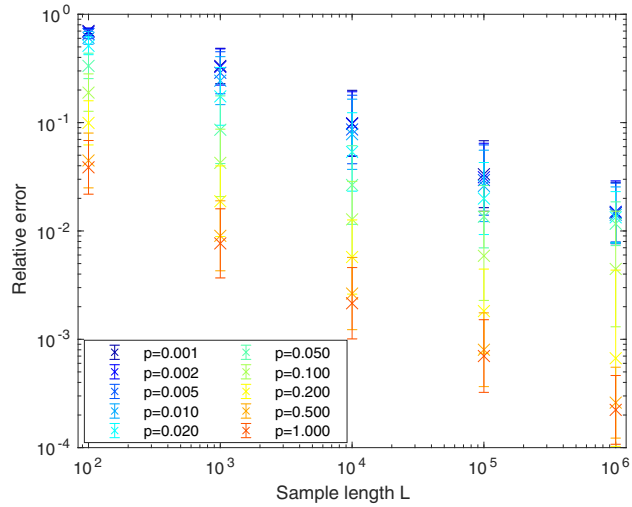

Fig. 1. Numerical results validating Eq. **15** for average steady-state distance from synchronization $\langle\sigma^2\rangle$ for continuous-time dynamics against empirical measurements, throughout a small-world transition on an $N = 100$ ring network with network randomization parameter p. Parameters for the network connectivity C are described in Section 2.C, with $d = 4$ and $c = 0.5$. The plot shows the relative absolute error of the empirical measurements $\langle\sigma^2\rangle_E$, averaged over 2,000 network realizations, versus the number of time series samples L for the empirical measurements for each network. Error bars represent the SD across realizations in log space.

Crucially, this demonstrates that the analytic results are well validated against the empirical experiments because the empirical results converge exponentially to the expected analytic values as the number of samples (which $\langle\sigma^2\rangle_E$ is averaged over) increases. Moreover, this convergence is observed across all values of network randomization parameter p, from regular ring to small-world to random networks as p increases from 0 to 1. We also see that there is a smaller relative error for random networks for the same number of samples L; we will explain why this is the case after we analytically relate synchronizability to network structure in the next section.

We perform the same experiment validating the solution for the discrete-time dynamics in Eq. **2** in *SI Appendix*, Fig. S1, where the same conclusions hold.

3. Results: Relating Network Motifs to Synchronizability

In the spirit of Barnett et al.'s (45, 61) analysis of the Tononi–Sporns–Edelman (TSE) complexity (55) in networks (since inspiring or mirrored by studies of several other measures of dynamics on networks (42, 56, 57, 62, 63)), we now use the power-series expansions in Eqs. **15** and **16** to directly reveal the roles of motif structures in $\langle\sigma^2\rangle$.

First, we define the following notation. We use $\mathbf{w}_{a\to b,M}$ to represent the weighted walk motif count of all directed walks from node a to b of length M:

$$\begin{aligned}\mathbf{w}_{a\to b,M} &= \sum_{n_1,n_2\ldots n_{M-1}} C_{an_1}C_{n_1n_2}\ldots C_{n_{M-1}b},\\ &= (C^M)_{ab} = ((C^M)^T)_{ba},\end{aligned} \quad [17]$$

and for shorthand write C^M_{ab} for $(C^M)_{ab}$ as the (a, b) entry in C^M (as distinct from the edge weight C_{ab} raised to power M).

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We also use $\mathbf{w}_{a\to b,M_1}^{a\to e,M_2}$ to represent weighted dual walk motif counts as the product of the two weighted walk motif counts starting from node a and finishing at nodes b and e, respectively:

$$\begin{aligned}\mathbf{w}_{a\to b,M_1}^{a\to e,M_2} &= \mathbf{w}_{a\to b,M_1}\mathbf{w}_{a\to e,M_2},\\ &= (C^{M_1})_{ab}(C^{M_2})_{ae}, && [18]\\ &= ((C^{M_1})^T)_{ba}(C^{M_2})_{ae}. && [19]\end{aligned}$$

The subcase of $\mathbf{w}_{a\to b,M_1}^{a\to b,M_2}$ (i.e., where the walks start at node a and end at node b in both cases) are referred to as weighted closed dual walk motif counts, representing the weighted counts of pairs of walks starting at node a and ending at node b via walks of length M_1 and M_2, respectively.

By convention, $\mathbf{w}_{a\to b,0} := \delta_{ab}$, so $\mathbf{w}_{a\to b,0}^{a\to e,M_2} = \mathbf{w}_{a\to e,M_2}$ when $a = b$. Several examples of walk motifs included in these counts are shown in Figs. 2 and 3. Both $\mathbf{w}_{a\to b,M}$ and $\mathbf{w}_{a\to b,M_1}^{a\to e,M_2}$ are examples of *weighted process motif counts* (42), being counts of structured sets of walks on a network, and we note that many such process motifs or walks can occur on a given underlying structural motif (42).

Finally, note that our expansions will utilize the key result that the trace of a given matrix A after the centering projections are applied to produce $\mathrm{trace}(U^TAU)$ is given by

$$\mathrm{trace}(U^TAU) = \sum_i A_{ii} - \frac{1}{N}\sum_{i,j} A_{ij}, \qquad [20]$$

as proven in Lemma 5 in *SI Appendix*, section 2.

A. Continuous-Time Processes. Now, expanding Eq. **15** using Eq. **20**, the steady-state distance from sync becomes:

$$\langle\sigma^2\rangle = \frac{\zeta^2}{2\theta}\sum_{m=0}^{\infty}\frac{2^{-m}}{N}\sum_{u=0}^{m}\binom{m}{u} \times\left(\sum_i((C^u)^TC^{m-u})_{ii} - \frac{1}{N}\sum_{i,j}((C^u)^TC^{m-u})_{ij}\right),$$

so,

$$\langle\sigma^2\rangle = \frac{\zeta^2}{2\theta}\sum_{m=0}^{\infty}\frac{2^{-m}}{N}\sum_{u=0}^{m}\binom{m}{u} \times\left(\sum_{i,k}C_{ki}^uC_{ki}^{m-u} - \frac{1}{N}\sum_{i,j,k}C_{ki}^uC_{kj}^{m-u}\right), \qquad [21]$$

$$\langle\sigma^2\rangle = \frac{\zeta^2}{2\theta}\sum_{m=0}^{\infty}\frac{2^{-m}}{N}\sum_{u=0}^{m}\binom{m}{u} \times\sum_{i,k}\left(\mathbf{w}_{k\to i,u}^{k\to i,m-u} - \frac{1}{N}\sum_j\mathbf{w}_{k\to i,u}^{k\to j,m-u}\right), \qquad [22]$$

using Eq. **19**.

We see that the steady-state distance from synchronization $\langle\sigma^2\rangle$ is determined by weighted dual walk motif counts, as the difference between contributions of

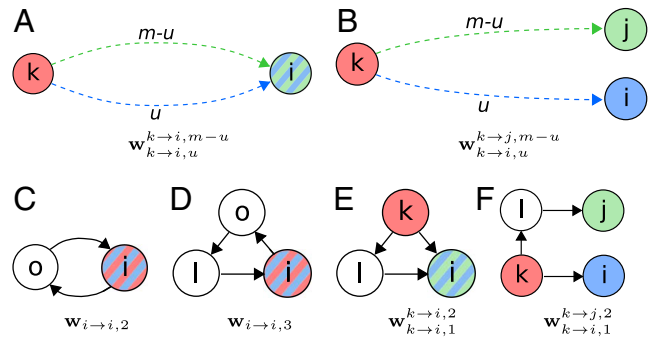

Fig. 2. Process motifs implicated in calculation of $\langle\sigma^2\rangle$ for the continuous-time process, being the difference between weighted counts of (*A*) closed dual walk motifs $\mathbf{w}_{k\to i,u}^{k\to i,m-u}$ (with two walks from node k to i in $m-u$ and u steps, respectively) and (*B*) all dual walk motifs $\mathbf{w}_{k\to i,u}^{k\to j,m-u}$ (with a walk from node k to i in u steps and a walk to j in $m-u$ steps, respectively) averaged over j. Samples of process motifs these involve include feedback loops (*C*) and (*D*), feedforward loops (*E*), and general open dual walk motifs where $j \neq i$ (*F*), as well as all of those shown for the discrete-time process in Fig. 3. Subfigure labels indicate the weighted counts to which the displayed process motifs contribute. Nodes are colored as per k, i, and j for $\mathbf{w}_{k\to i,u}^{k\to j,m-u}$ in (*B*), with cross-hatching on nodes indicating node $k = i$ in (*C*) and (*D*), and node $j = i$ in (*A*) and (*E*).

1. closed dual walk motif counts $\mathbf{w}_{k\to i,u}^{k\to i,m-u}$, being in general feedforward loop motifs from source nodes k to targets i in lengths u and $m-u$, including the subcases of
   (a) feedback loop motifs via $\mathbf{w}_{i\to i,u}^{i\to i,m-u}$ (where $k = i$) as the motif of two feedback loops on nodes i of cycle lengths u and $m-u$ (or a single feedback loops $\mathbf{w}_{i\to i,m}$ of cycle length m when $u = 0$ or m); and
2. all dual walk motif counts $\mathbf{w}_{k\to i,u}^{k\to j,m-u}$ from sources k to nodes i and j over (potentially different) lengths u and $m-u$, averaged over all j.

Some sample low-order closed and other dual walk motifs contributing here to $\langle\sigma^2\rangle$ are shown in Fig. 2. We will discuss the interpretation of this result in Section 4.

B. Discrete-Time Processes. Expanding Eq. **16** in a similar way to the continuous process (see full derivation in *SI Appendix*, section 5), we obtain

$$\langle\sigma^2\rangle = \frac{\zeta^2}{N}\sum_{u=0}^{\infty}\left(\sum_{i,k}\mathbf{w}_{k\to i,u}^{k\to i,u} - \frac{1}{N}\sum_{i,j,k}\mathbf{w}_{k\to i,u}^{k\to j,u}\right). \qquad [23]$$

We see that $\langle\sigma^2\rangle$ here is determined similarly from weighted dual walk motif counts, this time as the difference between contributions of

1. closed dual walk motif counts $\mathbf{w}_{k\to i,u}^{k\to i,u}$ from source nodes k to targets i in two walks of the same length u, including the subcases of
   (a) feedback loop motifs via $\mathbf{w}_{i\to i,u}^{i\to i,u}$ (where $k = i$) as the motif of two feedback loops on node i of the same cycle lengths $u \geq 1$; and
2. all dual walk motif counts $\mathbf{w}_{k\to i,u}^{k\to j,u}$ from sources k to nodes i and j over the same length u, averaged over all j.

Samples of process motifs contributing to the lowest-order closed and other dual walk motif counts relevant to $\langle\sigma^2\rangle$ here are shown in Fig. 3.

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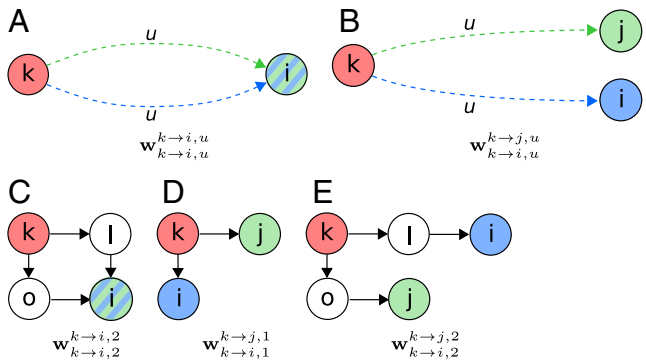

Fig. 3. Process motifs implicated in calculation of $\langle\sigma^2\rangle$ for the discrete-time process, being the difference between weighted counts of (*A*) closed dual walk motifs $\mathbf{w}_{k\to i,u}^{k\to i,u}$ and (*B*) averages of all dual walk motifs $\mathbf{w}_{k\to i,u}^{k\to j,u}$ (with the dual walks of the same length u in both cases). Samples of process motifs these involve include: the feedback loops shown in Fig. 2, feedforward loops of the same length u (*C*), and general open dual walk motifs of the same length u where $j \neq i$ (*D*) and (*E*). Subfigure labels indicate the weighted counts to which the displayed process motifs contribute. Nodes are colored as per k, i, and j for $\mathbf{w}_{k\to i,u}^{k\to j,u}$ in (*B*), with cross-hatching on nodes indicating node $j = i$ in (*A*) and (*C*).

4. Further Analysis and Discussion

A. Relative Prevalence of Closed Dual Walk Process Motifs Determines Synchronizability. While the precise process motifs involved in determining synchronizability for continuous- and discrete-time processes are subtly different in Eqs. **22** and **23**, the overall story is quite similar. Synchronizability is reduced (with increasing $\langle\sigma^2\rangle$) as the weighted count of closed dual walk motifs increases compared to what may be expected from all dual walk motifs ending at any two nodes (not necessarily ending at the same node). We can phrase this in several ways. For node pairs k and i connected by a walk of length u, Eq. **22** can be interpreted as estimating the difference between the actual amount of (weighted) redundancy from k to i of length $m-u$ (captured in $\mathbf{w}_{k\to i,u}^{k\to i,m-u}$) versus the amount expected if the walks of length $m-u$ from k to i were weighted as walks from k to all other nodes j on average (captured in $\frac{1}{N}\sum_j \mathbf{w}_{k\to i,u}^{k\to j,m-u}$). The more such redundancy in our walks, the less synchronizable the network becomes (with higher $\langle\sigma^2\rangle$). Put another way, it could be seen as a weighted consideration of whether walks of length $m-u$ from node k—which has a walk to i in u steps—preferentially end back at node i, as compared to such walks ending at another node j of the network.

Indeed, this is much easier to interpret if we assume for a moment that all edge weights are positive and of constant magnitude and compare networks of fixed degree. In this case, the interpretation becomes largely a numbers game, and we see immediately that synchronizability is reduced ($\langle\sigma^2\rangle$ is increased) as more walks from nodes k to i become closed by other directed walks from k to i (rather than those walks reaching some other node j). Here then, the more feedback and feedforward process motifs or loops we have in our network (for discrete time only closed process motifs of the same length), or in other words, the more clustered structure we have, the worse the synchronizability of the network becomes.

B. Detrimental Effects of Clustered Structure. The above provides a direct explanation for the result that maximally synchronizable networks have no directed loops (41), and moreover, the widely observed result (1, 9, 26, 28, 30–32, 39) that random networks are more synchronizable than small-world or regular ring networks. In other words, because small-world or regular networks simply have more feedback and feed-forward process motifs in their more clustered structure, they exhibit worse synchronizability in general. This result applies directly where edge weights are positive, but can also be expected to be similar when the edge weights are positive on average, as is commonly the case in biological networks such as in the mammalian cortex (61).

We numerically explore this idea in Fig. 4, continuing our experiment on sample network structures throughout a small-world transition on an $N = 100$ Watts–Strogatz ring network, in continuous-time with $\theta = \zeta = 1$, with various in-degrees d and total incoming connection weights c as outlined in Section 2.C. Again, Matlab code for these experiments is distributed in the linsync toolbox (58). Fig. 4 *A*–*E*, for a range of different combinations of in-degree d and connection weight c, each directly shows a strong transition in the steady-state distance from synchronization $\langle\sigma^2\rangle$, changing from being very large (i.e., relatively poor synchronization) for regular ring network structures (low p) through to a substantial proportion smaller (i.e., relatively good synchronization) for random network structures (high p). The relative synchronizability improves rapidly through the small-world regime; e.g., for $d = 4$, $c = 0.5$ (Fig. 4*B*) we observe $\langle\sigma^2\rangle$ dropping by around 25% with only 1% of links randomized, and by a factor of 3 by the time 10% of links are randomized.

We also see much less variance in $\langle\sigma^2\rangle$ across network samples once 5 to 10% of the edges are randomized since the networks then become more homogeneous across nodes and therefore across network samples as well. (The initial large increase in variance for $d = 2$, $c = 0.5$ in Fig. 4*A* is discussed in *SI Appendix*, section 7). This greater homogeneity across nodes also leads to lower variance in dynamics, which would explain the smaller relative error of empirical results against the analytic values of $\langle\sigma^2\rangle$ in Fig. 1.

The result that regular and small-world networks exhibit worse synchronization in general is subtle and is not just a numbers game though—we emphasize it is about the weighted difference of closed versus all dual walk process motifs. More clustered structure will lead to worse synchronizability under the above constraints of positive constant magnitude edge weights and fixed degree but does not necessarily mean worse synchronizability when relevant aspects of the network are not well controlled, as is made clear by Arenas et al. (1) and Nishikawa and Motter (21). For example, increasing the degree of nodes in a small-world network will lead to higher counts of loop process motifs, but eventually, this leads to a fully connected network which is known to have optimal synchronizability (20).** As above, one needs to consider not only numbers of such process motifs but the relative difference between closed and all such counts: In the fully connected network case, there may be many closed dual walk process motifs, but there are many more that remain unclosed. Indeed, this provides a direct argument against extrapolating from studies of synchronizability of motifs in isolation (36–38) because the relative proportions become very skewed compared to what they may look like when embedded in a network.

**We can already see in Fig. 4 *A*–*C* that increasing the in-degree substantially drops $\langle\sigma^2\rangle$, to a point that for $d = 8$ with $p \to 0$, it indicates a lower distance from synchronization than for substantially more randomized networks with $d = 2$. This would be confusing if one thought only about closed walk process motif counts, without thinking about the difference between closed and open and properly controlling relevant aspects of the networks.

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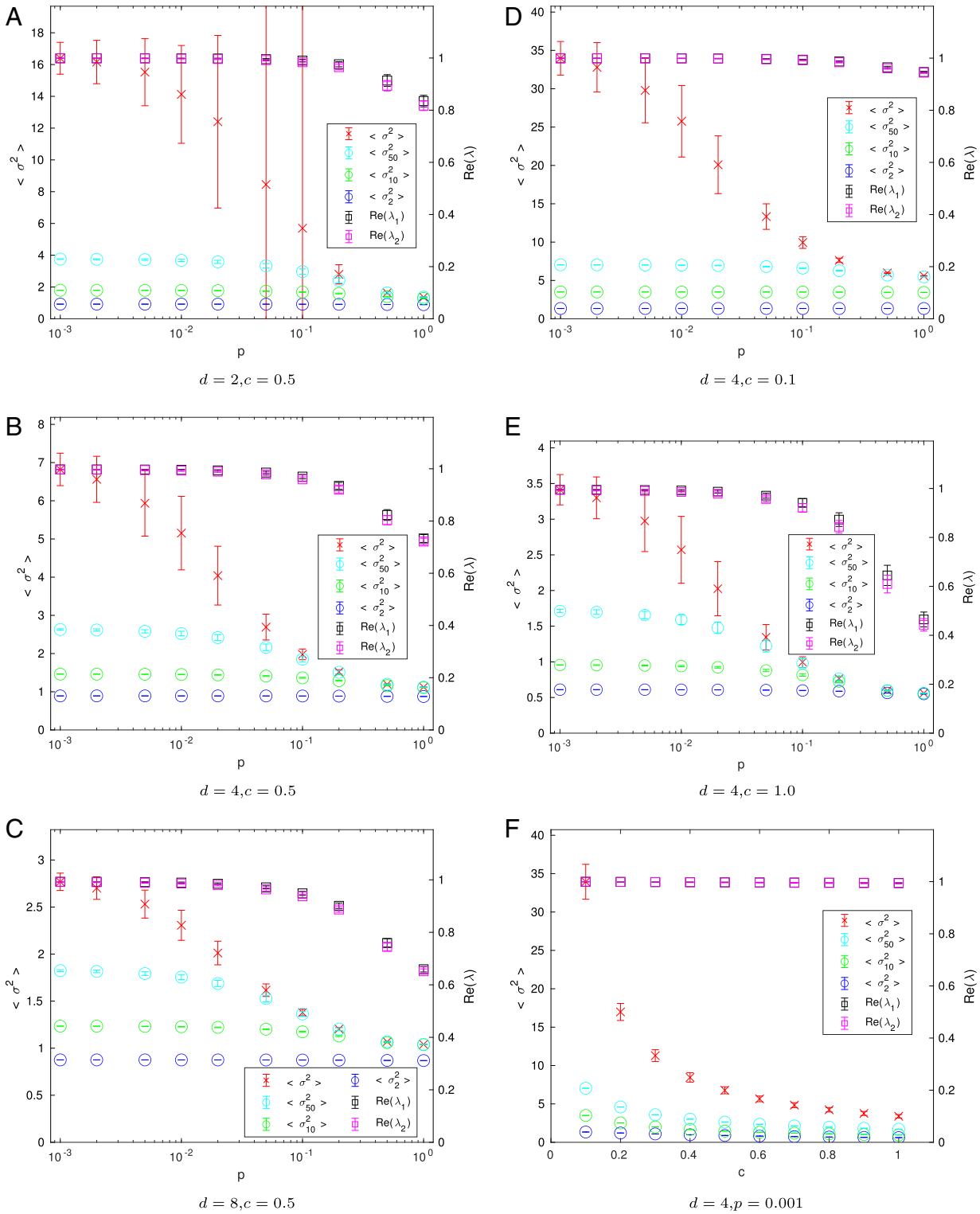

Fig. 4. Numerical results for synchronization throughout a small-world transition for continuous-time dynamics on an $N = 100$ Watts–Strogatz ring network. Parameters for the network connectivity matrix are described in Section 2.C, including in-degree d, total incoming (nonself) connection weight c, and network randomization parameter p, (e.g., in (*A*)–(*E*) regular ring networks appear on the *left* as $p \to 0$ and random networks on the *right* as $p \to 1$). Various measures and heuristics for the quality of synchronization computed analytically from the network structure alone (without simulating dynamics) are plotted, including the average steady-state distance from synchronization $\langle \sigma^2 \rangle$ computed via our method in Eq. **15** (presented in Section 2.B), and low-order approximations $\langle \sigma^2_M \rangle$ (defined in Eq. **24** in Section 4.E), and the largest two real components of the eigenvalues of C (except for ψ_0, discussed in Section 4.H). The mean value for each measure for each parameter combination is taken over 2,000 network realizations, and error bars represent the SD. *SI Appendix*, Fig. S2 reproduces (*A*) showing the full extent of the error bars.

C. Interpretation in Terms of Perturbation Reinforcement. We can also interpret these results in terms of resilience to perturbations on node k in Eqs. **22** and **23**. Closed dual walks from k to i (again considering positive edge weights only) lead to such perturbations eventually reinforcing on node i, driving it away from synchronizing with other nodes (as observed for

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variance for the same dynamics by Schwarze and Porter (42)). In contrast, a greater proportion of walks to other nodes j will dissipate perturbations more evenly through the network and thus disturb synchronization to a lesser degree.

We can also look at this in terms of competition between clusters for what the dominant value to synchronize around will be: more clustering leads to larger perturbation reinforcement within the cluster and stronger competition between the clusters, which harms overall global synchronization. Indeed, this illustrates the known result that increasing synchronizabilty within two largely separate network clusters can harm synchronizabilty of the overall network (52). From an information-theoretic perspective, loop process motif structures increase information transfer within such clustered modules (63), reinforcing synchronization within clusters at the expense of synchronizabilty of the overall network.

D. Contrasting Continuous- and Discrete-Time Results. Indeed, this interpretation in terms of perturbations allows us to see why the contributing motifs are different for continuous- and discrete-time processes. For discrete-time processes, perturbations on node k will only reinforce on node i from two walks of the same length since the driving noise $\vec{r}(t)$ is uncorrelated in time. In continuous-time processes, considered to be more realistic for modeling biological processes (45), nodes in contrast experience an uncorrelated rate $d\vec{w}(t)$ of perturbations, though their integration over time means that perturbations on node k can be reinforced on node i from two walks of different lengths. Hence, more general closed dual walk motifs contribute to synchronizability for continuous-time processes, including feedforward process motifs with different length walks. As we see from the $\binom{m}{u}$ term in Eq. **22**, the more similar the lengths of the dual walks of length u and $m-u$ are, the stronger their contribution (for fixed m). This was also observed for covariance for the same dynamics (42), since this ensures a maximal reinforcement of the perturbation.

E. Stronger Contribution of Shorter Walks. Shorter walks contribute more strongly to the expressions for $\langle\sigma^2\rangle$ for these dynamics, as has been observed for other measures (42, 45, 62) since under the stability of the dynamics (orthogonal to ψ_0), information is gradually "forgotten" along the relevant walks.

We can investigate this numerically by defining low-order approximations $\langle\sigma_M^2\rangle$ to $\langle\sigma^2\rangle$, for motif sizes up to an order M. In continuous-time, M represents the longest total length of the two dual walks $(m-u, u)$ from k to i considered:

$$\langle\sigma_M^2\rangle = \frac{\zeta^2}{2\theta}\sum_{m=0}^{M}\frac{2^{-m}}{N}\sum_{u=0}^{m}\binom{m}{u} \times \sum_{i,k}\left(\mathbf{w}_{k\to i,u}^{k\to i,m-u} - \frac{1}{N}\sum_{j}\mathbf{w}_{k\to i,u}^{k\to j,m-u}\right), \qquad [24]$$

while in discrete time, M is more naturally defined as the longest length of each of the dual walks of length u from k to i that is considered:

$$\langle\sigma_M^2\rangle = \frac{\zeta^2}{N}\sum_{u=0}^{M}\left(\sum_{i,k}\mathbf{w}_{k\to i,u}^{k\to i,u} - \frac{1}{N}\sum_{i,j,k}\mathbf{w}_{k\to i,u}^{k\to j,u}\right). \qquad [25]$$

These low-order approximations $\langle\sigma_M^2\rangle$ for $M = \{2, 10, 50\}$ are plotted for the various numerical experiments in Fig. 4, and in all cases, we see that the low-order approximations for $M = \{2, 10\}$ account for a proportionally larger component of $\langle\sigma^2\rangle$ than the higher-order contributions that would also be included in $\langle\sigma_{50}^2\rangle$. This directly shows that the shorter walks contribute more strongly than the longer walks. This effect is particularly striking for more random networks ($p \to 1$), where the low-order approximations are already very close to the true value of $\langle\sigma^2\rangle$. This is because the nontrivial amount of clustered structure in the more regular networks induces many more convergent walks, particularly at longer walk lengths, which need to be accounted for. That shorter walks have stronger contribution in the more random networks ($p \to 1$) is less clear for low cross-coupling cases such as $c = 0.1$ in Fig. 4*D*; there, the more heavily weighted self-links ($C_{ii} = 1 - c = 0.9$) are not randomized with p and dominate the walk counts, meaning that higher-order approximations of $\langle\sigma_M^2\rangle$ are required to tease out the differences with p.

As the coupling strengths are increased though, the longer walks begin to have larger impacts on $\langle\sigma^2\rangle$. This effect was explored for other measures by refs. 45 and 61. We can see directly from Eq. **22** that network-wide increases in coupling will nonlinearly increase $\langle\sigma^2\rangle$, until reaching a point where $\langle\sigma^2\rangle$ diverges (meaning the network is not synchronizable). This also reflects the result that any network structure can be made synchronizable by lowering the magnitudes of the C_{ji} (17). Illustrating the increased impact of longer walks with coupling strength is somewhat complicated in the numerical experiments here though, because increasing the cross-coupling c necessarily means reducing the self-coupling $C_{ii} = 1 - c$ (required to maintain ψ_0), and both contribute to $\langle\sigma^2\rangle$. Indeed, Fig. 4*F* shows that $\langle\sigma^2\rangle$ actually decreases as the cross-coupling c increases as observed by ref. 20 (although the relative proportion accounted for by the longer walks increases)—but again, this is solely because of the related increase in self-coupling $C_{ii} = 1 - c$ in our model, which dominates the changes to $\langle\sigma^2\rangle$ there. A more clear illustration emerges if we vary in-degree $d = \{2, 4, 8\}$ for fixed total incoming weights $c = 0.5$ and therefore fixed self-coupling $C_{ii} = 1 - c = 0.5$ in Fig. 4 *A*–*C*: Lowering the in-degree increases the weights on each nonself edge, and although this means that there are less cross-edges and less shorter walks, their larger weights lead to a substantial increase in $\langle\sigma^2\rangle$ away from synchronization, and a larger influence of the longer walks in that increase.

F. Lower Synchronizability for Symmetric Networks. Aside from direct connections contributing via $\mathbf{w}_{k\to i,1}^{k\to i,1}$, the shortest and therefore strongest contributing closed dual walk process motifs are recurrent connections $\mathbf{w}_{i\to i,1}^{i\to i,1}$ and feedback loops of length 2 which are involved, e.g., in $\mathbf{w}_{i\to i,2}^{i\to i,u}$.

The latter are of particular interest since in undirected or symmetric networks C reciprocal connections—feedback loops of length 2—occur for every connected pair of nodes. In comparison to directed networks with the same fixed in-degree, and fixed positive edge weights (or majority positive as above), this will make substantial contributions to increasing $\langle\sigma^2\rangle$ in Eqs. **22** and **23**, and thereby reducing synchronizability. In other words, this explains why undirected networks are generally less synchronizable than corresponding directed networks (as observed, e.g., in ref. 64).

Considering the $\mathbf{w}_{i\to i,2}^{i\to i,2}$ as one such term acting to increase $\langle\sigma^2\rangle$ for example, it is easily demonstrated for fixed positive

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edge weights that the only way for that term to be completely countered by the corresponding $\frac{1}{N}\sum_j \mathbf{w}_{i\to i,2}^{i\to j,2}$ term would be for each outgoing neighbor of i to connect onward to every node in the network. (Which of course, reflects the aforementioned situation in full connectivity where closed dual walk process motifs lose any sense of being comparatively common.)

Another perspective on this result is that symmetric connectivity is known to increase autocorrelation and slow down dynamics (65). From an information-theoretic perspective, symmetric connectivity and loop process motif structures in general are known to increase self-predictability of nodes embedded in a network, as measured by active information storage (62), and higher active information storage has been observed to be negatively correlated to synchronizability (66). Such self-predictability may indicate that a node is less responsive to the dynamics of others in the network, and therefore not synchronizing as well with them.

G. Inhibition versus Excitation. Much of the above intuition comes from considering positive or excitatory edge weights only, or majority positive edge weights, and is generally flagged as such. We must however be aware that the presence of negative or inhibitory edge weights can have dramatic effects on our interpretation of the results here. The mathematics of Eqs. **22** and **23** are unchanged; we simply need to remember that the weighted walk counts incorporate these negative edge weights. This has been emphasized particularly by Nishikawa and Motter (21) and is very relevant for this study.

As an example, consider the reciprocally connected pairs referred to within the term $\mathbf{w}_{i\to i,2}^{i\to i,u}$ above as reducing synchronizability—for positive edge weights. Where one edge weight in the pair is negative, this switches the sign of the contribution and therefore leads to the relevant term actually enhancing synchronization—assuming that all of the other weights in the other feedback loop $\mathbf{w}_{i\to i,u}$ as part of this term are positive. This illustrates the difficulty in extending our intuition to the presence of negative edge weights since multiple instances of them can be quite confusing.

H. Relation to Previous Approaches. We emphasize the generality of our result in comparison to previous work in not requiring symmetry, nonnegativity, or diagonalizability of the weighted coupling matrix C. This is important since real networks are often directional and weighted and since optimal synchronization requires nondiagonalizable networks (41). Indeed, nondiagonalizable matrices C are encountered even with our very simple model for generating Watts–Strogatz ring networks for the numerical experiments in Fig. 4, e.g., being a substantial proportion of networks for $d = 2$, $c = 0.5$ from $P \geq 0.05$ and becoming the majority for larger P.

While extremal eigenvalues determine whether sync is possible at all, as they control the fastest and the slowest decaying modes, Eq. **10** here emphasizes that fully characterizing the relative synchronizability requires taking all of the eigenvalues of C into account (as echoed by ref. 50). This contrasts with common heuristics for relative synchronizability which focus on the extremal eigenvalues or their spread (as per Section 1.B). The common heuristic for continuous-time dynamics is that synchronization is improved as $\mathrm{Re}(\lambda_1)$, the largest real component of an eigenvalue of C (aside from $\lambda_0 = 1$ corresponding to ψ_0) moves further away from the stability boundary of 1; or in other words, synchronization improves as $\mathrm{Re}(\lambda_1)$ decreases.[††] To investigate how well this heuristic performs, $\mathrm{Re}(\lambda_1)$ is plotted in the results of our numerical experiments in Fig. 4. Each subplot—across a wide range of parameters—shows that this heuristic provides a remarkably different indication of how the quality of synchronization changes with respect to p and c, as compared to the ground truth now provided by $\langle\sigma^2\rangle$. In particular, we see in Fig. 4 *A–E* that while changes in $\langle\sigma^2\rangle$ toward improved synchronization are visible well before network randomization parameter P reaches 0.01 (or on average changes to 4 out of 400 edges here), the heuristic of $\mathrm{Re}(\lambda_1)$ requires an order of magnitude larger randomization (p around 0.1) before any change is visible. This contrast is more stark in the experiment varying cross-coupling strength in Fig. 4*F*, where there is no visible change at all in the heuristic $\mathrm{Re}(\lambda_1)$, despite the underlying distance from synchronization $\langle\sigma^2\rangle$ indicating an order of magnitude improvement. The second-largest real component $\mathrm{Re}(\lambda_2)$ is also included in these plots, with similar results. Moreover, with the applicability of $\langle\sigma^2\rangle$ to deterministic systems and linearizations of nonlinear systems demonstrated in *SI Appendix*, sections 1 and 6, it has no less scope for use than heuristic eigenvalue analysis. This result underlines the importance of using our precise measure of synchronizability rather than the common heuristics from the extremal eigenvalues.

There are other interesting mathematical connections from our result to previous work. *SI Appendix*, section 7 (c.f. Eq. **94**) shows for instance how the result for continuous time in Eq. **22** simplifies for symmetric networks with $\theta = \zeta = 1$ to give the existing result Eq. **10** by Korniss (20) for these in terms of inverse sums of eigenvalues, as it should. We also contribute a similar simplification for symmetric networks in discrete time with $\zeta = 1$ (derived in *SI Appendix*, section 7):

$$\langle\sigma^2\rangle = \frac{1}{N}\sum_{\lambda\neq\lambda_0}\frac{1}{1-\lambda_C^2}. \qquad [26]$$

While the two continuous-time results are equivalent when $|\lambda_v| < 1$ (except for λ_0), our power series result does not converge outside this range (within $\mathrm{Re}(\lambda_1) < 1$) even though $\langle\sigma^2\rangle$ has a solution for symmetric networks from Eq. **10**; future work may establish how our solution could be adapted to provide further insight into these conditions.

Further insights using Eq. **10** for symmetric networks with $\theta = \zeta = 1$ are found by interpreting C as the transition matrix for random walks on a network (with $C_{ji} \geq 0$ being probabilities when $\vec{\psi}_0$ is an eigenvector with eigenvalue $\lambda_0 = 1$). Here then, we can immediately observe that the steady-state distance from synchronization $\langle\sigma^2\rangle$ for C in continuous-time in Eq. **10** is directly related to Kemeny's constant $K(C)$ (67, 68), as $\langle\sigma^2\rangle = \frac{K(C)}{2N}$. Kemeny's constant captures the (weighted) average travel times of random walks between two nodes in the network and can also be interpreted in terms of how well connected the nodes are, and as an indication of clustering (69). Kemeny's constant also directly maps to the "effective graph resistance" or Kirchhoff index for regular graphs (70, 71), which is interpreted as the difficulty of transport in the network, and so, we see that $\langle\sigma^2\rangle$ also directly maps to this quantity in this case. These interpretations are consistent with the insights our power series formalism brings for $\langle\sigma^2\rangle$, e.g., the relations to clustering. Moreover, the insights our formalism brings in terms

[††]This is complementary to the expression of synchronization improving as $\mathrm{Re}(\lambda_{L1})$ for the smallest eigenvalue of L increases (52).

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of the relative abundance of convergent process motifs are thus directly applicable for Kemeny's constant and effective graph resistance and will find further utility in those studies. This is particularly important for Kemeny's constant, for example, where gaining intuition into the measure has been an ongoing concern (67, 69). Similarly, our process motif interpretation could be usefully applied to the use of effective graph resistance to study the robustness of complex networks, which has known applications in mitigating cascading failures in electrical networks (71). Finally, we note that most work on Kemeny's constant has considered undirected networks (68, 69, 71), and as such, our directed process motif interpretation may contribute to studies on directed networks (72, 73) here.

Another interesting connection is found in combining Eqs. **9** and **20**, giving

$$\langle \sigma^2 \rangle = \frac{1}{N} \sum_i \Omega_{ii} - \frac{1}{N^2} \sum_{i,j} \Omega_{ij}. \qquad [27]$$

This directly shows us that $\langle \sigma^2 \rangle$ is a difference between the average autocovariance of all nodes and the average covariance between all node pairs (including the autocovariances). Crucially, this difference converges under the stated conditions in Section 2.A, even though the various (auto)covariance terms may not (i.e., where ψ_0 is an eigenvector).

Indeed, this difference maps directly back to the difference between closed and all dual walk motif counts in our results: The weighted counts for dual walk motifs converging on node i determine the autocovariance for node i, while the weighted counts of dual walk motifs terminating at nodes i and j determine their covariance. That these types of process motifs determine the autocovariance and covariance was previously observed by Schwarze and Porter (42), and they necessarily recur here in determining synchronizability as above.

We also note that the mathematics of the dual walk motif counts in Eqs. **22** and **23** resemble the power series expansion of the exponential of the adjacency matrix in the communicability function (74) (albeit with different weightings for the contributions of different walk lengths), with the terms forming the difference in Eq. **27** themselves resembling self- and nonself-communicability (75).

I. Biological Implications. While our results consider walks on networks, or process motifs, rather than directly counting structural motifs, we know that increasing counts of structural motifs with closed paths will lead to increasing counts of the closed walk motifs observed here (42) (specifically where edge weights are positive on average, as observed earlier, being commonly the case in biological networks such as in the mammalian cortex (61)). And indeed, such closed-loop structural motifs are observed in many real-world networks.

A particularly interesting example, where synchronization is of special relevance, lies in brain networks. The high level of modularity in brain networks—which is particularly important in compartmentalizing function for cognition, and in conserving wiring length (76)—also serves to increase counts of closed-loop motifs and therefore push the brain away from global synchronization. It is interesting to note that synapses of bidirectionally connected pairs of neurons are on average stronger than synapses of unidirectionally connected pairs (65), which will also specifically act against global synchronization. These results are important since synchronization is associated with pathological conditions such as epilepsy and Parkinson's disease (1). Regional synchronization within a module in contrast may be useful and could be enhanced by more local connectivity within modules.

5. Conclusion

In this paper, we have provided a fully general analytic calculation of synchronizability which requires neither symmetric nor diagonalizable connectivity matrix C. This is an important theoretical advance in and of itself, allowing us to handle network structures which are more commonly found and to compute the steady-state distance from synchronization $\langle \sigma^2 \rangle$ without such restrictions. Moreover, our formalism is applicable to considering linearization around attractors in nonlinear dynamics and synchronization therein, and we have demonstrated the equivalence of the $\langle \sigma^2 \rangle$ measure of steady-state deviation from synchronization in noise-driven dynamics with total summed deviation from synchronization due to single perturbations in deterministic dynamics. Overall, our work provides a substantial advance over heuristic approximations from the eigenvalue spectra (e.g., examining leading eigenvalues and the spread of the spectrum) that had been previously in use, allowing us to properly compute the synchronizability for a given network structure here.

Perhaps more importantly, we are able to interpret our result in terms of weighted directed walks on networks, or process motifs. Indeed, this allows us to fully comprehend the role of network structure in determining synchronizability. Along these lines, we can now completely explain several well-known results, including that regular ring and small-world networks tend to be further from synchronization than random networks, and the manner in which clustered structure leads to reduced global synchronization.

These results are important more generally in characterizing the relationship between network structure and dynamics or function, since addressing this for synchronization has historically been seen as one of the most important canonical problems in this field (1, 12).

Along these lines, the results here will have applicability beyond the apparent scope of the Ornstein–Uhlenbeck or Edwards–Wilkinson process and vector-autoregressive dynamics, with the former having been observed as "entirely equivalent" to a linearized Wilson–Cowan noisy neural network (45) and is also similar to covariances of spike rates in systems of coupled Hawkes processes (56, 57).

There is much scope for further extension of the results included here. Primarily, the exact results for the steady-state distance from synchronization provided here should be used to systematically investigate the plethora of insights that have been gained in the past using heuristics from the eigenspectra—as we have shown here such heuristics can easily miss very drastic changes in the synchronization characteristics as we alter network properties. Future work will also involve investigating extensions to handle time delays in the coupling between nodes in the network in the models in Eqs. **1** and **2**. The case of time delays requires nontrivial extensions to the mathematics here, and it will be interesting to see how such time delays change the general solutions and provide interpretability of changes in synchronizability of a given network.

Data, Materials, and Software Availability. Code data have been deposited in Github (https://github.com/jlizier/linsync) (58).

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